Machine Learning - CA2.1

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For this CA, I wish to look at Housing Prices I selected the following Dataset from kaggle:

https://www.kaggle.com/datasets/yasserh/housing-prices-dataset

I will split this data into training and test data

First I wish to create a Baseline Multi-Linear Regression Model and evaluate it's performance, using House Price as the Dependant Variable

Afterwards I would like to transform the independant variables using principle component analysis and conduct another regression on the transformed data On this model I wish to use Grid Search to tune the number of PCA components to use in the second multi-linear regression model

And if I have enough time, I would like to try out a third model, random forest regressor, to compare between the three models to see which is strongest at predicting house prices

I will evaluate performance of each based on their R and R-Squared and Mean Squared Error values aswell as other metrics.

This code will also be uploaded to my github repo here:

https://github.com/Matthew-Riddell/Machine-Learning-CA2.1

Question 1: Import data:

Import relevant libraries and load the dataset.

Importing Packages

In [248…

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
import statsmodels.api as sm
import scipy
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split, GridSearchCV
```

```
from sklearn.metrics import r2_score, mean_squared_error
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
```

Loading the dataset

In [202…

```
filename = 'Housing.csv'
house_data = pd.read_csv(filename)
```

In [203…

```
house_data.head()
```

Out[203…

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basem
0	13300000	7420	4	2	3	yes	no	
1	12250000	8960	4	4	4	yes	no	
2	12250000	9960	3	2	2	yes	no	
3	12215000	7500	4	2	2	yes	no	
4	11410000	7420	4	1	2	yes	yes	

Data Cleaning and Preparation

Replacing the yes / no values with 1 / 0

In [204…

```
house_data.replace({'yes':1, 'no':0}, inplace=True)
```

```
C:\Users\Matty\AppData\Local\Temp\ipykernel_12876\1967606657.py:1: FutureWarnin
g: Downcasting behavior in `replace` is deprecated and will be removed in a fut
ure version. To retain the old behavior, explicitly call `result.infer_object
s(copy=False)`. To opt-in to the future behavior, set `pd.set_option('future.n
o_silent_downcasting', True)`
  house_data.replace({'yes':1, 'no':0}, inplace=True)
```

Dropping furnishing status variabke

In [205…

```
house_data = house_data.drop("furnishingstatus", axis=1)
```

Checking for missing values

In [206…

```
print("Number of missing values in each column:")
print(house_data.isna().sum())
```

```
Number of missing values in each column:
price              0
area               0
bedrooms           0
bathrooms          0
stories            0
mainroad           0
guestroom          0
basement           0
hotwaterheating    0
airconditioning    0
parking            0
prefarea           0
dtype: int64
```

In [207…

```python
total_missing = house_data.isna().sum().sum()
print("Total missing values in the dataset:", total_missing)
```

```
Total missing values in the dataset: 0
```

Checking for nonsense values

In [208…

```python
nonsense_values = ["?", "error", "missing", "Missing", "NaN", "nan", "N/A", "r
```

Checks every column for any of the values in the nonsense values list

In [209…

```python
print("Checking for nonsense values:")
for col in house_data.columns:
    count = house_data[col].isin(nonsense_values).sum()
    if count > 0:
        print(col, "has", count, "nonsense values")
    else:
        print("No nonsense values found!")
```

```
Checking for nonsense values:
No nonsense values found!
No nonsense values found!
No nonsense values found!
No nonsense values found!
No nonsense values found!
No nonsense values found!
No nonsense values found!
No nonsense values found!
No nonsense values found!
No nonsense values found!
No nonsense values found!
No nonsense values found!
```

In [210…

```python
house_data.head()
```

Out[210…

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basem
0	13300000	7420	4	2	3	1	0	
1	12250000	8960	4	4	4	1	0	
2	12250000	9960	3	2	2	1	0	
3	12215000	7500	4	2	2	1	0	
4	11410000	7420	4	1	2	1	1	

There are no missing values or nonsense values in the dataset, data is clean, converted yes / no into binary values and dropped the furnishing status variable

Question 2: Data Description:

Display data attributes using data visualization libraries/modules.

You must use text, code, and visualization tools and comment on Data features suitable for analysis.

Checking how many columns and rows the dataset has

In [211…

```
house_data.shape
```

Out[211…

```
(545, 12)
```

Displaying the each of the variables and their object types

In [212…

```
house_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 545 entries, 0 to 544
Data columns (total 12 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   price             545 non-null    int64
 1   area              545 non-null    int64
 2   bedrooms          545 non-null    int64
 3   bathrooms         545 non-null    int64
 4   stories           545 non-null    int64
 5   mainroad          545 non-null    int64
 6   guestroom         545 non-null    int64
 7   basement          545 non-null    int64
 8   hotwaterheating   545 non-null    int64
 9   airconditioning   545 non-null    int64
 10  parking           545 non-null    int64
 11  prefarea          545 non-null    int64
dtypes: int64(12)
memory usage: 51.2 KB
```

The dataset I am using has the following variables:

area: the area of the house in square feet

bedrooms: the number of bedrooms in the house

bathrooms: the number of bathrooms in the house

stories: the number of stories the house has

mainroad: yes/no if the house is connected to a main road

guestroom: yes/no if the house has a guest room

basement: yes/no if the house has a basement

hotwaterheating: yes/no if the house has a hot water heater

airconditioning: yes/no if the house has air conditioning

parking: number of parking spaces the house has

prefarea: yes/no if the house is in a preferred area

In [213…

```
house_data.head(15)
```

Out[213…

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	bas
0	13300000	7420	4	2	3	1	0	
1	12250000	8960	4	4	4	1	0	
2	12250000	9960	3	2	2	1	0	
3	12215000	7500	4	2	2	1	0	
4	11410000	7420	4	1	2	1	1	
5	10850000	7500	3	3	1	1	0	
6	10150000	8580	4	3	4	1	0	
7	10150000	16200	5	3	2	1	0	
8	9870000	8100	4	1	2	1	1	
9	9800000	5750	3	2	4	1	1	
10	9800000	13200	3	1	2	1	0	
11	9681000	6000	4	3	2	1	1	
12	9310000	6550	4	2	2	1	0	
13	9240000	3500	4	2	2	1	0	
14	9240000	7800	3	2	2	1	0	

Summary statistics of each variable

In [214…

```
house_data.describe()
```

Out[214…

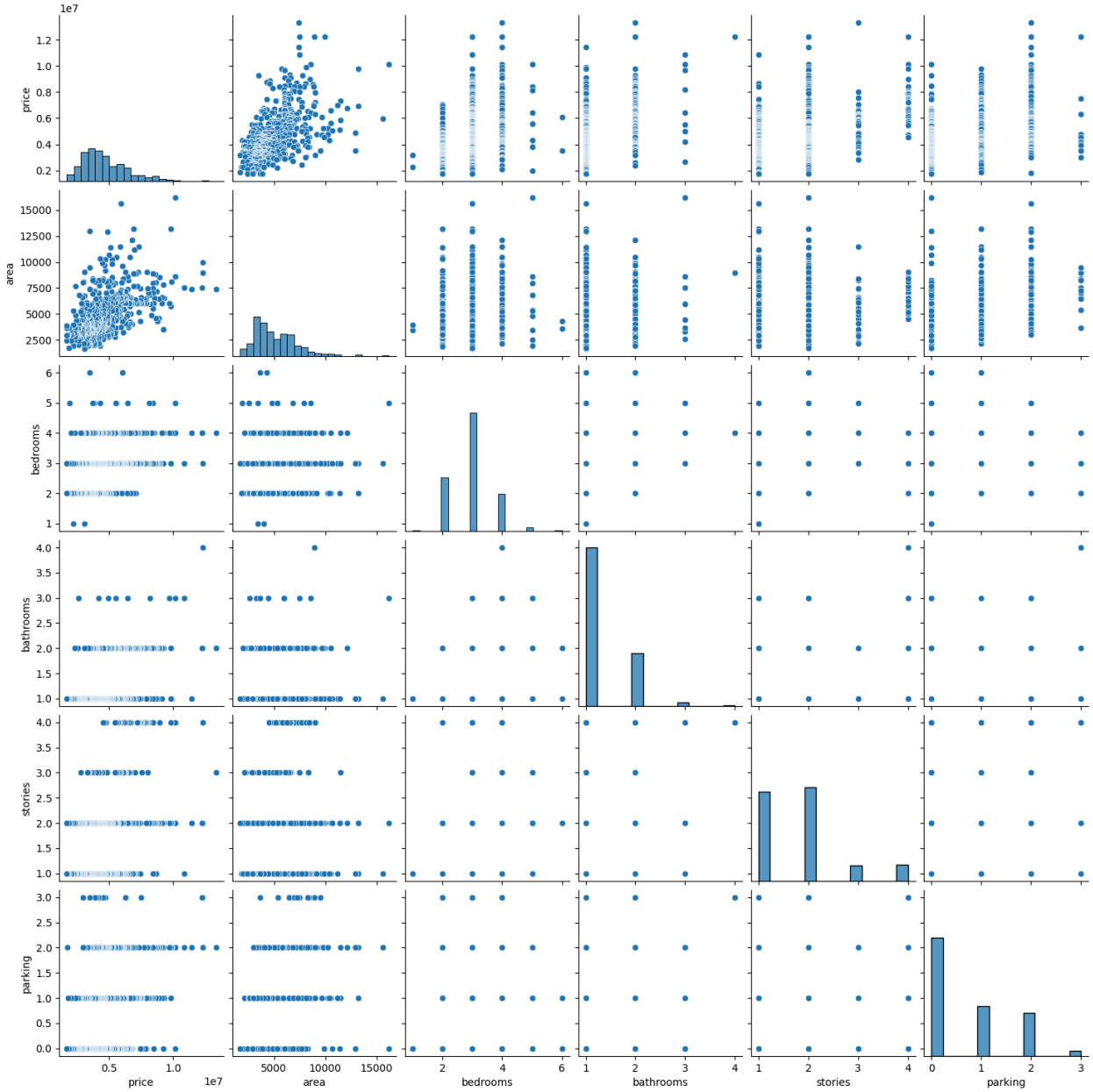
	price	area	bedrooms	bathrooms	stories	mainro
count	5.450000e+02	545.000000	545.000000	545.000000	545.000000	545.0000
mean	4.766729e+06	5150.541284	2.965138	1.286239	1.805505	0.8587
std	1.870440e+06	2170.141023	0.738064	0.502470	0.867492	0.3486
min	1.750000e+06	1650.000000	1.000000	1.000000	1.000000	0.0000
25%	3.430000e+06	3600.000000	2.000000	1.000000	1.000000	1.0000
50%	4.340000e+06	4600.000000	3.000000	1.000000	2.000000	1.0000
75%	5.740000e+06	6360.000000	3.000000	2.000000	2.000000	1.0000
max	1.330000e+07	16200.000000	6.000000	4.000000	4.000000	1.0000

Scatterplots of each numerical variable

In [215…

```
num_cols = ['price', 'area', 'bedrooms', 'bathrooms', 'stories', 'parking']
```

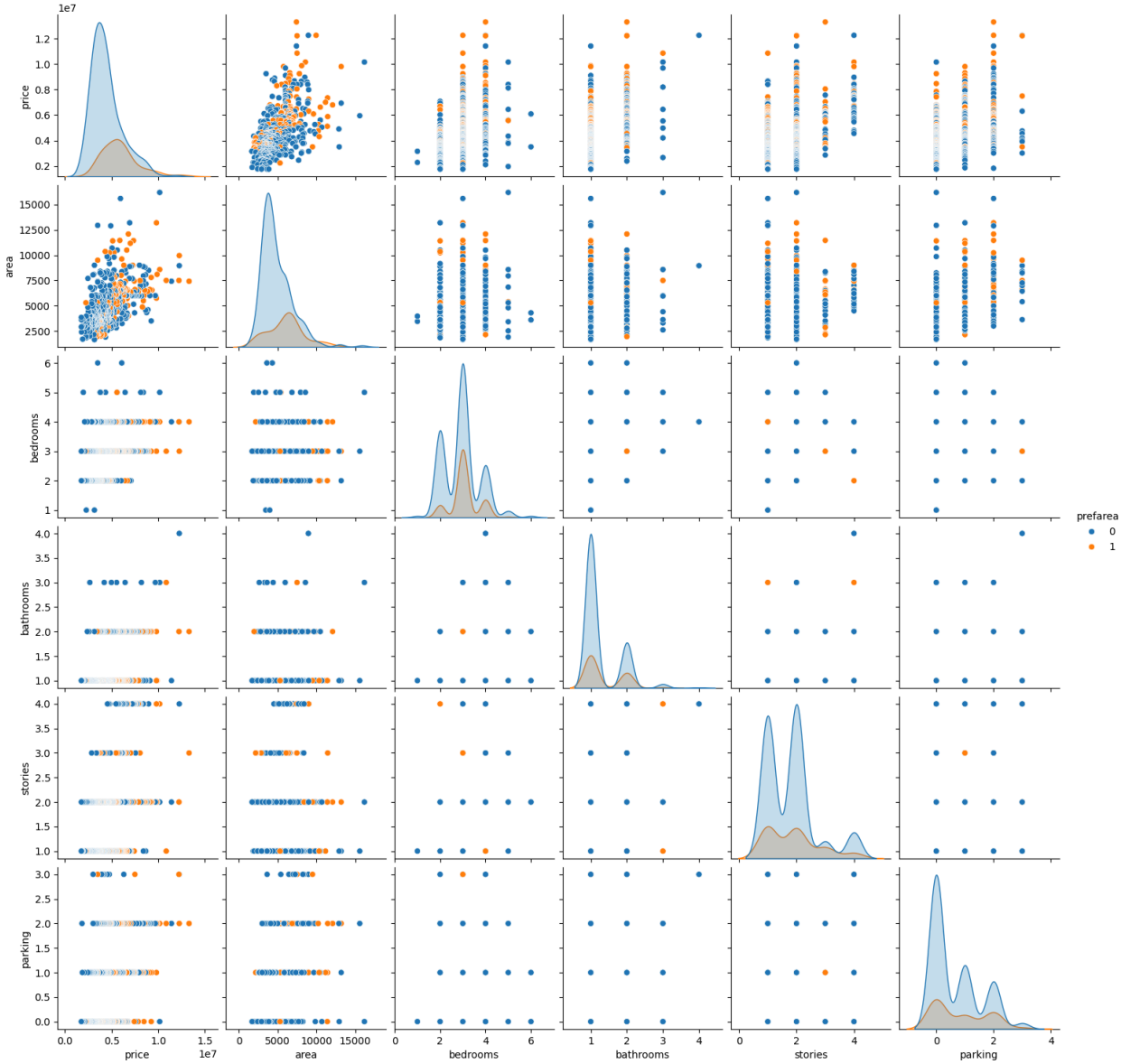
```
sns.pairplot(house_data[num_cols])
plt.show()
```



Scatterplots with prefarea highlighted

In [216…

```
sns.pairplot(house_data[num_cols + ['prefarea']], hue='prefarea')
plt.show()
```



Most of these variables appear to be non-linear

Area is strongly correlated with Price and Bedroom and Stories also have looser correlations

Pearson correlations between each numerical variable

In [217…

```
house_data[num_cols].corr()
```

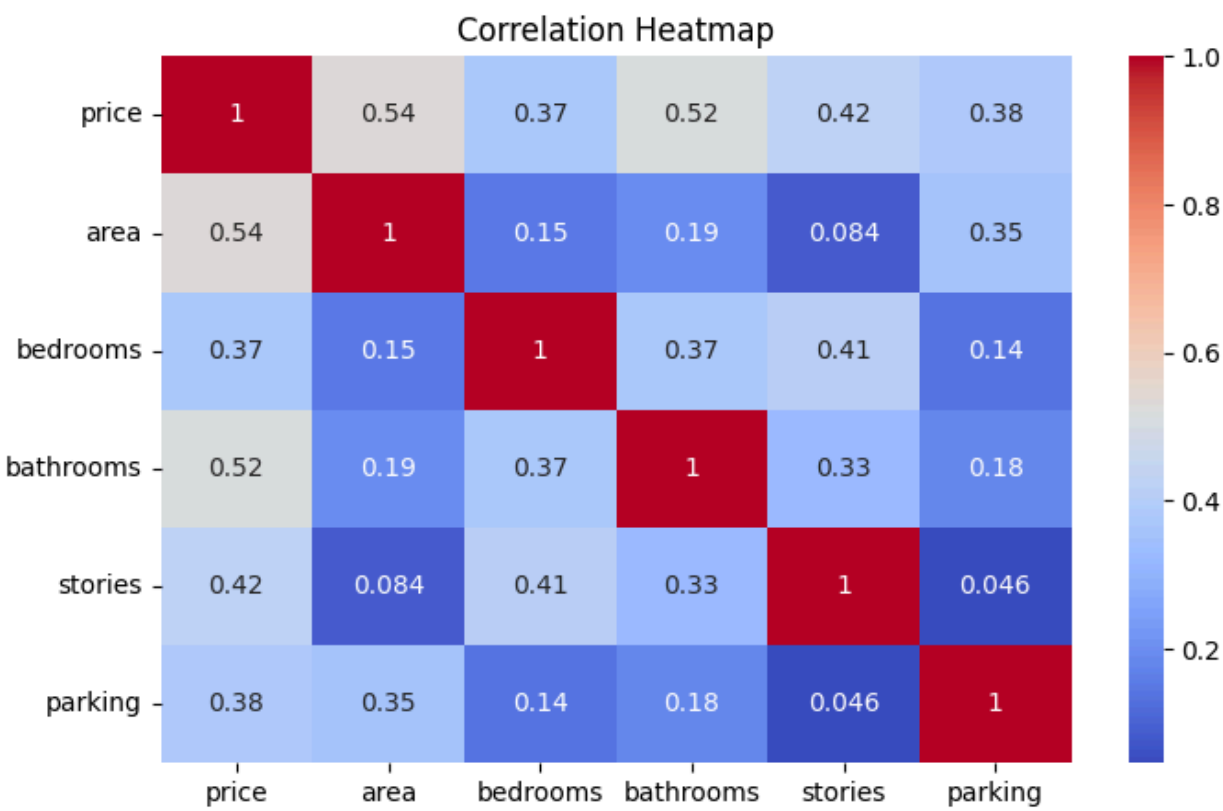
Out[217…

	price	area	bedrooms	bathrooms	stories	parking
price	1.000000	0.535997	0.366494	0.517545	0.420712	0.384394
area	0.535997	1.000000	0.151858	0.193820	0.083996	0.352980
bedrooms	0.366494	0.151858	1.000000	0.373930	0.408564	0.139270
bathrooms	0.517545	0.193820	0.373930	1.000000	0.326165	0.177496
stories	0.420712	0.083996	0.408564	0.326165	1.000000	0.045547
parking	0.384394	0.352980	0.139270	0.177496	0.045547	1.000000

Correlation heatmap

In [218…

```
plt.figure(figsize=(8,5))
sns.heatmap(house_data[num_cols].corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



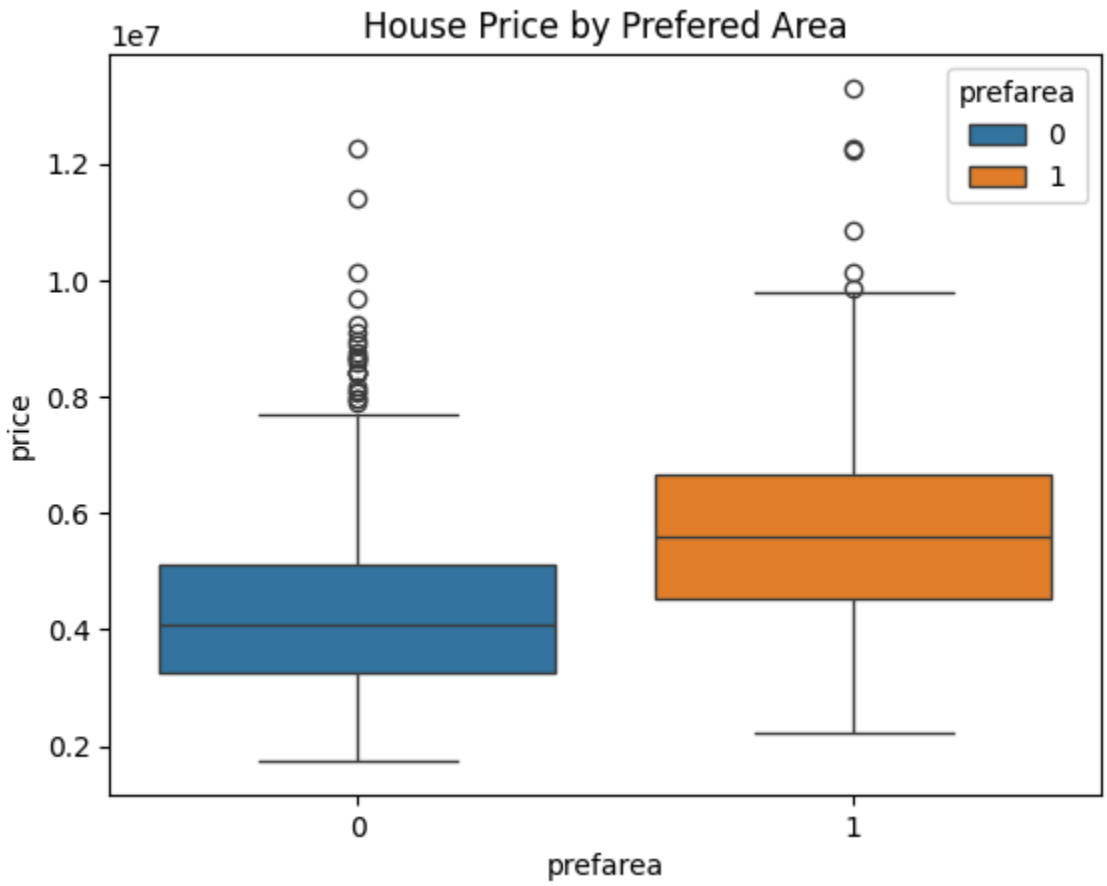
All the independant variables appear to be uncorrelated which is good news for avoiding multi-collinearity

Boxplots of house prices vs Prefered Area

In [219…

```
sns.boxplot(x='prefarea', y='price', data=house_data, hue='prefarea')
```

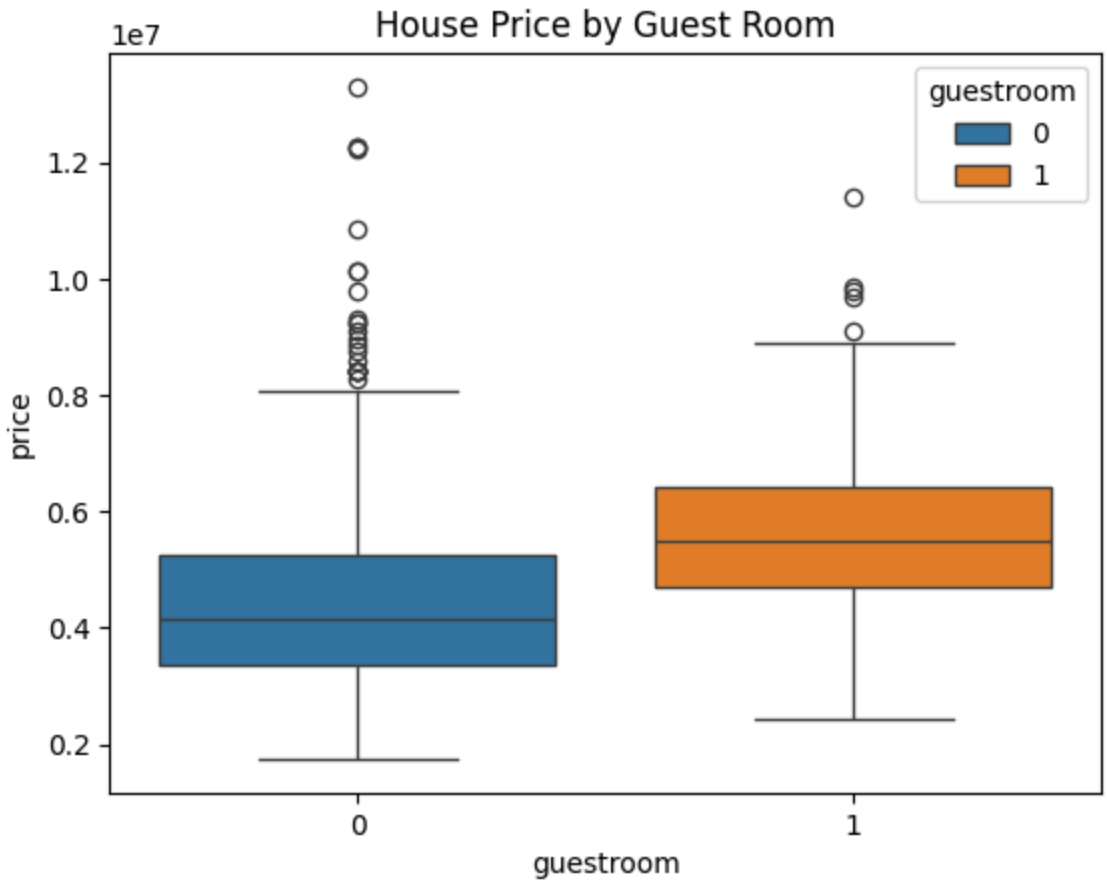
```
plt.title("House Price by Prefered Area")
plt.show()
```



Boxplots of house prices vs Guest Room

In [220…

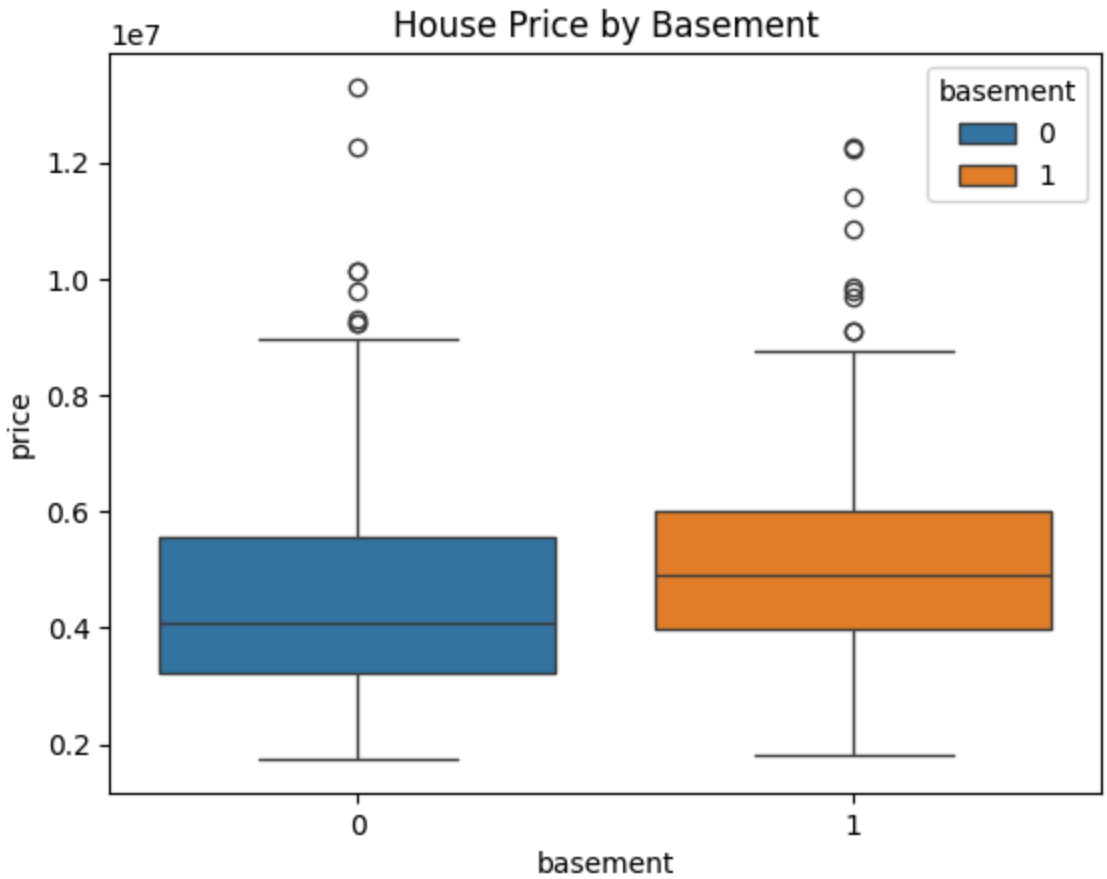
```
sns.boxplot(x='guestroom', y='price', data=house_data, hue='guestroom')
plt.title("House Price by Guest Room")
plt.show()
```



Boxplots of house prices vs Basement

In [221…

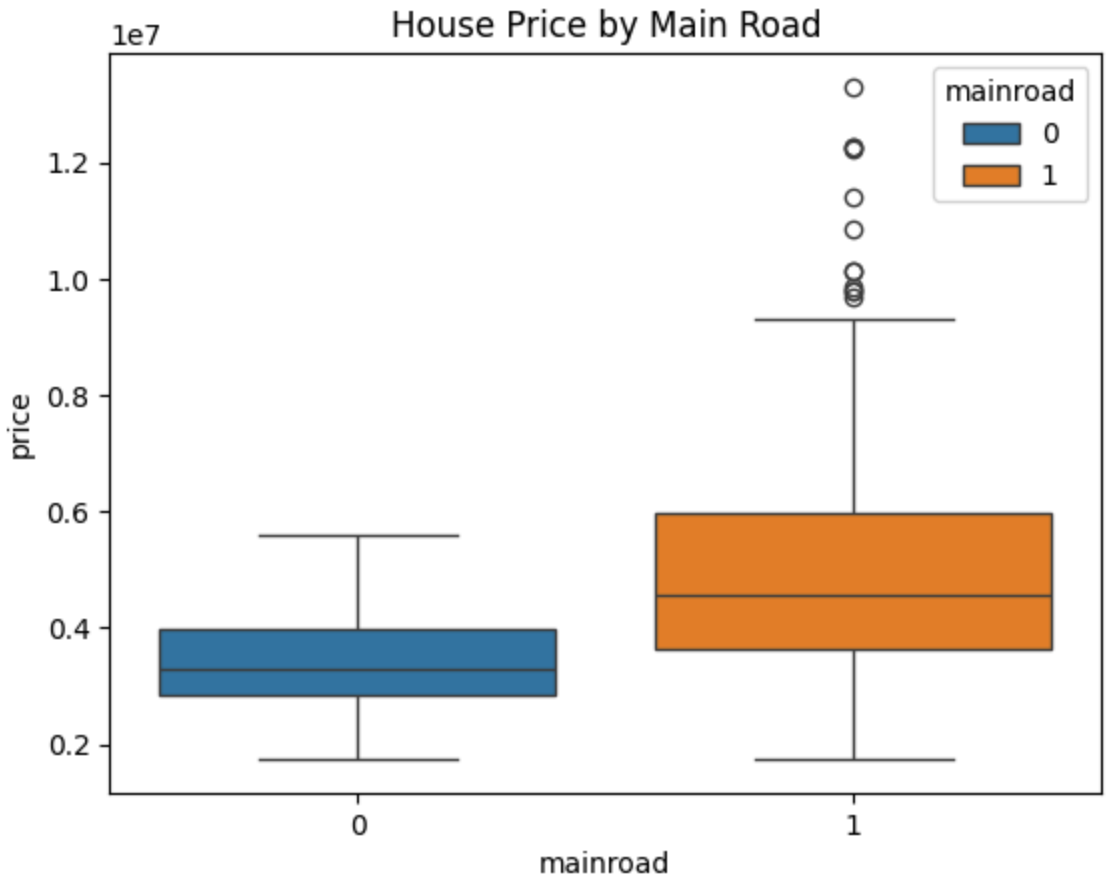
```
sns.boxplot(x='basement', y='price', data=house_data, hue='basement')
plt.title("House Price by Basement")
plt.show()
```



Boxplots of house prices vs Main Road

In [222…

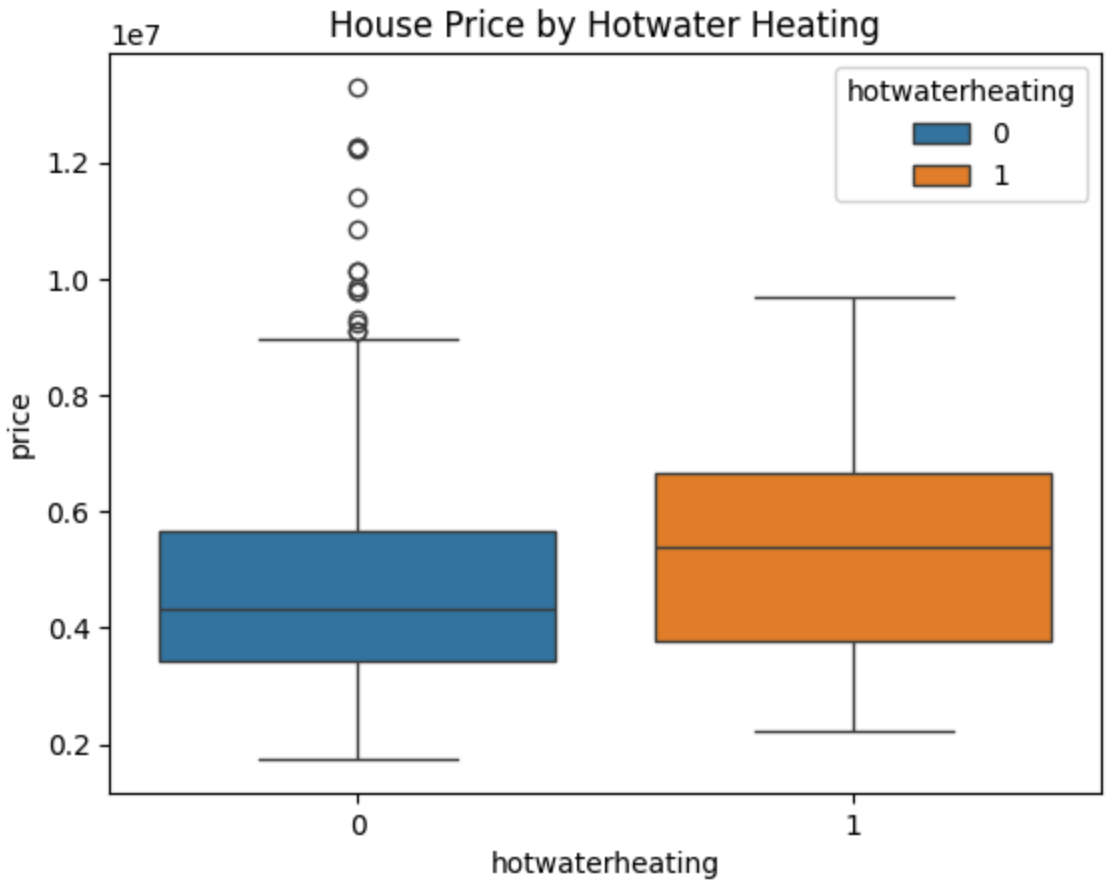
```
sns.boxplot(x='mainroad', y='price', data=house_data, hue='mainroad')
plt.title("House Price by Main Road")
plt.show()
```



Boxplots of house prices vs Hotwater Heating

In [223…

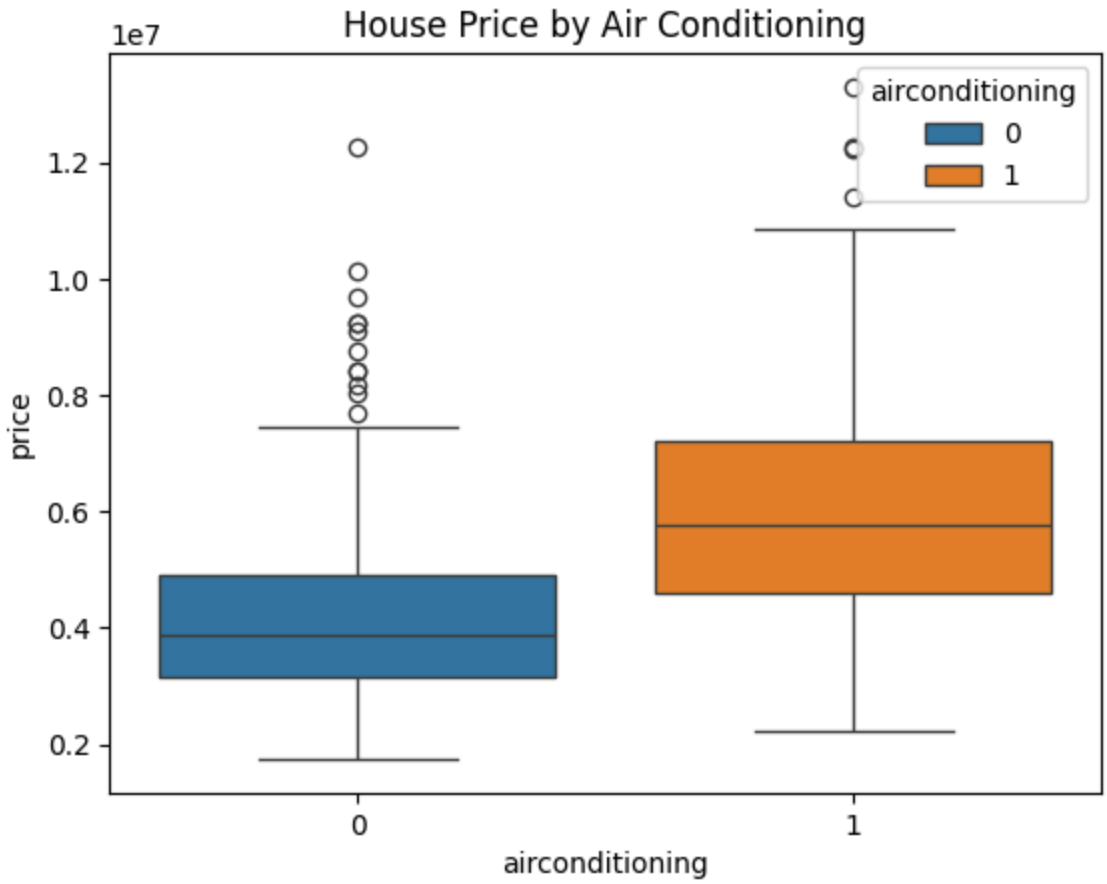
```
sns.boxplot(x='hotwaterheating', y='price', data=house_data, hue='hotwaterheat
plt.title("House Price by Hotwater Heating")
plt.show()
```



Boxplots of house prices vs Air Conditioning

In [224…

```
sns.boxplot(x='airconditioning', y='price', data=house_data, hue='airconditior
plt.title("House Price by Air Conditioning")
plt.show()
```



From looking at the boxplots

Each of the binary variables show an increase in price whenever the value is set to 1

The numerical variables have some correlations with but they are much weaker with area and number of bedrooms being the strongest

For the Multi-linear regression I may consider dropping some of the weakest correlated numerical and binary variables but for now I want to try it with all variables included

Question 3: ML Modeling:

Create / select a baseline ML model.

Determine the impact of selected ML model/models and interpret the best model with parameters.

Splitting Price into Y from the independant variables which remain as X

In [225…
```
X = house_data.drop('price', axis=1)
Y = house_data['price']
```

Splitting the data into training and test sets with a 20% split for the test data

In [226…
```
# Code adapted from week 4 parametric tuning activity
X_train, X_test, Y_train, Y_test = train_test_split(
    X, Y, test_size=0.2, random_state=42
)
```

Scaling the data, models such as PCA are highly sensitive to scaling

In [227…
```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Creating the linear regression model

fitting the model

In [228…
```
X_train_const = sm.add_constant(X_train)
model_lr = sm.OLS(Y_train, X_train_const).fit()
```

Predictions of Y based on the test data

In [229…
```
# Code adapted from week 4 parametric tuning activity
X_test_const = sm.add_constant(X_test)
Y_pred = model_lr.predict(X_test_const)
```

Model summary

In [230…
```
model_lr.summary()
```

Out[230…

OLS Regression Results

Dep. Variable:	price	**R-squared:**	0.678
Model:	OLS	**Adj. R-squared:**	0.670
Method:	Least Squares	**F-statistic:**	81.21
Date:	Tue, 17 Mar 2026	**Prob (F-statistic):**	4.48e-97
Time:	02:50:46	**Log-Likelihood:**	-6640.6
No. Observations:	436	**AIC:**	1.331e+04
Df Residuals:	424	**BIC:**	1.335e+04
Df Model:	11		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	-7.243e+04	2.47e+05	-0.293	0.770	-5.59e+05	4.14e+05
area	237.7105	25.144	9.454	0.000	188.288	287.133
bedrooms	7.848e+04	7.61e+04	1.031	0.303	-7.11e+04	2.28e+05
bathrooms	1.109e+06	1.15e+05	9.608	0.000	8.82e+05	1.34e+06
stories	4.269e+05	6.92e+04	6.170	0.000	2.91e+05	5.63e+05
mainroad	4.127e+05	1.5e+05	2.750	0.006	1.18e+05	7.08e+05
guestroom	2.436e+05	1.42e+05	1.716	0.087	-3.54e+04	5.23e+05
basement	4.35e+05	1.18e+05	3.690	0.000	2.03e+05	6.67e+05
hotwaterheating	7.123e+05	2.25e+05	3.165	0.002	2.7e+05	1.15e+06
airconditioning	8.084e+05	1.18e+05	6.876	0.000	5.77e+05	1.04e+06
parking	2.474e+05	6.2e+04	3.990	0.000	1.26e+05	3.69e+05
prefarea	6.368e+05	1.22e+05	5.211	0.000	3.97e+05	8.77e+05

Omnibus:	72.064	**Durbin-Watson:**	1.893
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	187.623
Skew:	0.810	**Prob(JB):**	1.81e-41
Kurtosis:	5.775	**Cond. No.**	3.05e+04

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.05e+04. This might indicate that there are strong multicollinearity or other numerical problems.

R-squared and RMSE

In [231…

```
# Code adapted from week 4 parametric tuning activity
r2_model_lr = r2_score(Y_test, Y_pred)
rmse_model_lr = np.sqrt(mean_squared_error(Y_test, Y_pred))
```

Displaying the R-squared and RMSE from the predictions

In [232…

```
print("Linear Regression R2:", r2_model_lr)
print("Linear Regression RMSE:", rmse_model_lr)
```

```
Linear Regression R2: 0.64372960866141
Linear Regression RMSE: 1341936.5557726547
```

Creating the PCA LInear Regression Model

using PCA to transform the training and testing data

In [233…

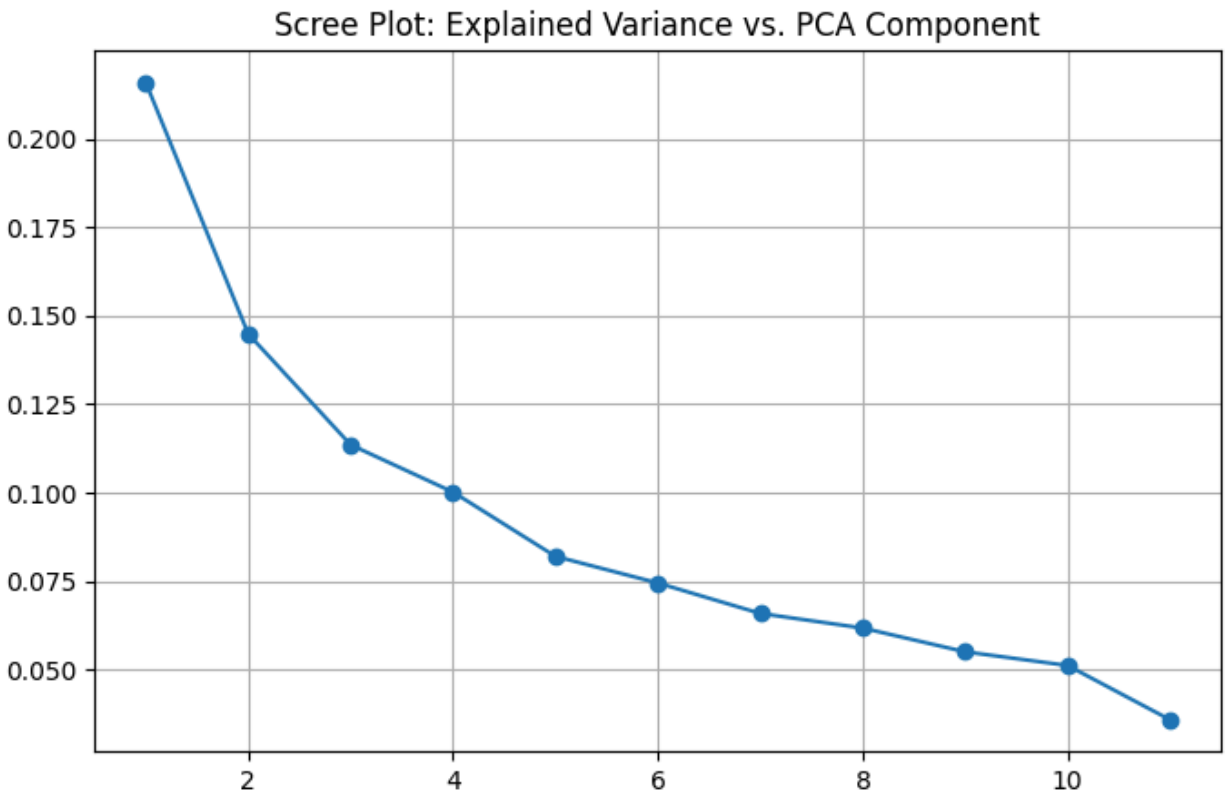
```
pca = PCA()

X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)
```

Scree Plot showing components and their explained variance

In [234…

```
plt.figure(figsize=(8, 5))
plt.plot(np.arange(1, len(X.columns) + 1), pca.explained_variance_ratio_, mark
plt.title('Scree Plot: Explained Variance vs. PCA Component')
plt.grid(True)
plt.show()
```



Using Grid Search for Hyper Parameter Tuning (finding the best set of PCA components)

In [235…

```
# Code adapted from week 4 parametric tuning activity
param_grid = {'n_components':[2,3,4,5,6,7,8]}
grid_search = GridSearchCV(estimator=pca, param_grid=param_grid, cv=5)
grid_search.fit(X_train_scaled)


print("Best number of components:", grid_search.best_params_)
```

```
Best number of components: {'n_components': 6}
```

Creating a new PCA with the tuned number of components

In [236…

```
model_pca = PCA(n_components=grid_search.best_params_['n_components'])

X_train_pca = model_pca.fit_transform(X_train_scaled)
X_test_pca = model_pca.transform(X_test_scaled)
```

Fitting the model

In [237…

```
X_train_pca_const = sm.add_constant(X_train_pca)
model_pca = sm.OLS(Y_train, X_train_pca_const).fit()
```

Predictions of Y based on the test data

In [238…

```
# Code adapted from week 4 parametric tuning activity
X_test_pca_const = sm.add_constant(X_test_pca)
Y_pred_pca = model_pca.predict(X_test_pca_const)
```

Model Summary

In [239…

```
model_pca.summary()
```

Out[239…

OLS Regression Results

Dep. Variable:	price	**R-squared:**	0.642
Model:	OLS	**Adj. R-squared:**	0.637
Method:	Least Squares	**F-statistic:**	128.5
Date:	Tue, 17 Mar 2026	**Prob (F-statistic):**	1.46e-92
Time:	02:50:46	**Log-Likelihood:**	-6663.5
No. Observations:	436	**AIC:**	1.334e+04
Df Residuals:	429	**BIC:**	1.337e+04
Df Model:	6		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	4.707e+06	5.07e+04	92.848	0.000	4.61e+06	4.81e+06
x1	9.069e+05	3.29e+04	27.577	0.000	8.42e+05	9.72e+05
x2	-2.177e+04	4.02e+04	-0.542	0.588	-1.01e+05	5.72e+04
x3	-7.065e+04	4.54e+04	-1.557	0.120	-1.6e+05	1.85e+04
x4	1.305e+05	4.83e+04	2.701	0.007	3.55e+04	2.25e+05
x5	-9439.5200	5.34e+04	-0.177	0.860	-1.14e+05	9.55e+04
x6	-3.821e+04	5.6e+04	-0.682	0.496	-1.48e+05	7.19e+04

Omnibus:	42.975	**Durbin-Watson:**	1.913
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	81.695
Skew:	0.585	**Prob(JB):**	1.82e-18
Kurtosis:	4.768	**Cond. No.**	1.70

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Display R-Squared and RMSE values of the model performance on test data

In [240…

```
# Code adapted from week 4 parametric tuning activity
r2_pca = r2_score(Y_test, Y_pred_pca)
rmse_pca = np.sqrt(mean_squared_error(Y_test, Y_pred_pca))
```

In [241…

```
print("PCA Model R2:", r2_pca)
print("PCA Model RMSE:", rmse_pca)
```

```
PCA Model R2: 0.6374186974488816
PCA Model RMSE: 1353769.7971816477
```

Creating a Random Forest Regressor Model

Fitting the model

In [260…

```
# Code adapted from week 4 parametric tuning activity
rf_model = RandomForestRegressor()
rf_model.fit(X_train, Y_train)
```

Out[260…

▾ RandomForestRegressor

Parameters

n_estimators	100
criterion	'squared_error'
max_depth	None
min_samples_split	2
min_samples_leaf	1
min_weight_fraction_leaf	0.0
max_features	1.0
max_leaf_nodes	None
min_impurity_decrease	0.0
bootstrap	True
oob_score	False
n_jobs	None
random_state	None
verbose	0
warm_start	False
ccp_alpha	0.0
max_samples	None
monotonic_cst	None

Predictions of Y based on the test data

In [261…
```
# Code adapted from week 4 parametric tuning activity
Y_pred_rf = rf_model.predict(X_test)
```

Displaying the R-Squared and RMSE values based on model performance on the test data

In [262…
```
# Code adapted from week 4 parametric tuning activity
r2_rf = r2_score(Y_test, Y_pred_rf)
rmse_rf = np.sqrt(mean_squared_error(Y_test, Y_pred_rf))
```

In [263…
```
print("Random Forest R2:", r2_rf)
print("Random Forest RMSE:", rmse_rf)
```

```
Random Forest R2: 0.61293485236197
Random Forest RMSE: 1398730.8456790478
```

Question 4: Evaluate the performance of the model in terms of evaluation metrics.

Provide a report on the performance comparison and explain the reasons for the following.

- Selection of model/models
- Selection of hyper-parameters
- Tuning of hyper-parameters
- Interpretation of the loss function (if any)
- Accuracy of your model during training and testing
- Submit the code with the final optimized model

Prepare a short report summarizing the steps taken to solve the problem.

Your report should clearly explain the approach, the models used, and the challenges encountered.

Focus on clarity and conciseness, and follow academic writing standards.

Evaluating the Multi-Linear Regression Model

Evaluation of the model on training data

In [245…
```
model_lr.summary()
```

Out[245…

OLS Regression Results

Dep. Variable:	price	**R-squared:**	0.678
Model:	OLS	**Adj. R-squared:**	0.670
Method:	Least Squares	**F-statistic:**	81.21
Date:	Tue, 17 Mar 2026	**Prob (F-statistic):**	4.48e-97
Time:	14:23:54	**Log-Likelihood:**	-6640.6
No. Observations:	436	**AIC:**	1.331e+04
Df Residuals:	424	**BIC:**	1.335e+04
Df Model:	11		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	-7.243e+04	2.47e+05	-0.293	0.770	-5.59e+05	4.14e+05
area	237.7105	25.144	9.454	0.000	188.288	287.133
bedrooms	7.848e+04	7.61e+04	1.031	0.303	-7.11e+04	2.28e+05
bathrooms	1.109e+06	1.15e+05	9.608	0.000	8.82e+05	1.34e+06
stories	4.269e+05	6.92e+04	6.170	0.000	2.91e+05	5.63e+05
mainroad	4.127e+05	1.5e+05	2.750	0.006	1.18e+05	7.08e+05
guestroom	2.436e+05	1.42e+05	1.716	0.087	-3.54e+04	5.23e+05
basement	4.35e+05	1.18e+05	3.690	0.000	2.03e+05	6.67e+05
hotwaterheating	7.123e+05	2.25e+05	3.165	0.002	2.7e+05	1.15e+06
airconditioning	8.084e+05	1.18e+05	6.876	0.000	5.77e+05	1.04e+06
parking	2.474e+05	6.2e+04	3.990	0.000	1.26e+05	3.69e+05
prefarea	6.368e+05	1.22e+05	5.211	0.000	3.97e+05	8.77e+05

Omnibus:	72.064	**Durbin-Watson:**	1.893
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	187.623
Skew:	0.810	**Prob(JB):**	1.81e-41
Kurtosis:	5.775	**Cond. No.**	3.05e+04

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.05e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Evaluation of the model's performance on training data

R-squared = 0.678 meaning that the model explains 67.8% of variance in Price

R-squared ajusted is almost identical at 0.670 confirming no overfitting of the model

P-value for co-efficent Area is extremely low (< 0.05) showing that it is highly significant

P-value for co-efficent Bedrooms is very high at 0.303 suggesting that it is a redundant, insignificant variable

P-value for co-efficent Bathrooms is extremely low (< 0.05) showing that it is highly significant

P-value for co-efficent Stories is extremely low (< 0.05) showing that it is highly significant

P-value for co-efficent Mainroad is low (< 0.05) showing that it is highly significant

P-value for co-efficent Guestroom is high (> 0.05) showing that it is somewhat redundant

P-value for co-efficent Basement is extremely low (< 0.05) showing that it is highly significant

P-value for co-efficent Hotwaterheating is extremely low (< 0.05) showing that it is highly significant

P-value for co-efficent Airconditioning is extremely low (< 0.05) showing that it is highly significant

P-value for co-efficent Parking is extremely low (< 0.05) showing that it is highly significant

P-value for co-efficent Prefarea is extremely low (< 0.05) showing that it is highly significant

Evaluating the PCA Transformed Multi-Linear Regression Model

Evaluation of the model on training data

In [246…

```
model_pca.summary()
```

Out[246…

OLS Regression Results

Dep. Variable:	price	**R-squared:**	0.642
Model:	OLS	**Adj. R-squared:**	0.637
Method:	Least Squares	**F-statistic:**	128.5
Date:	Tue, 17 Mar 2026	**Prob (F-statistic):**	1.46e-92
Time:	14:24:02	**Log-Likelihood:**	-6663.5
No. Observations:	436	**AIC:**	1.334e+04
Df Residuals:	429	**BIC:**	1.337e+04
Df Model:	6		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	4.707e+06	5.07e+04	92.848	0.000	4.61e+06	4.81e+06
x1	9.069e+05	3.29e+04	27.577	0.000	8.42e+05	9.72e+05
x2	-2.177e+04	4.02e+04	-0.542	0.588	-1.01e+05	5.72e+04
x3	-7.065e+04	4.54e+04	-1.557	0.120	-1.6e+05	1.85e+04
x4	1.305e+05	4.83e+04	2.701	0.007	3.55e+04	2.25e+05
x5	-9439.5200	5.34e+04	-0.177	0.860	-1.14e+05	9.55e+04
x6	-3.821e+04	5.6e+04	-0.682	0.496	-1.48e+05	7.19e+04

Omnibus:	42.975	**Durbin-Watson:**	1.913
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	81.695
Skew:	0.585	**Prob(JB):**	1.82e-18
Kurtosis:	4.768	**Cond. No.**	1.70

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Evaluation of the model's performance on training data

R-squared = 0.642 meaning that the model explains 64.2% of variance in Price

R-squared ajusted is almost identical at 0.637 confirming no overfitting of the model

P-value for co-efficent PC1 is extremely low (< 0.05) showing that it is highly significant

P-value for co-efficent PC2 is very high at 0.588 suggesting that it is a redundant, insignificant variable

P-value for co-efficent PC3 is high at 0.120 showing that it is redundant

P-value for co-efficent PC4 is extremely low (< 0.05) showing that it is highly significant

P-value for co-efficent PC5 is very high at 0.860 suggesting that it is a redundant, insignificant variable

P-value for co-efficent PC6 is very high at 0.496 suggesting that it is a redundant, insignificant variable

Random Forest Test Results:

- R-Squared 0.613
- RMSE 1398730.85

The model explains 61.3% of variance

RMSE very high

Evaluating all three models based on the performance from the test data

In [264…

```
print("Linear Regression R2:", r2_model_lr)
print("Linear Regression RMSE:", rmse_model_lr)

print("PCA Model R2:", r2_pca)
print("PCA Model RMSE:", rmse_pca)

print("Random Forest R2:", r2_rf)
print("Random Forest RMSE:", rmse_rf)
```

```
Linear Regression R2: 0.64372960866141
Linear Regression RMSE: 1341936.5557726547
PCA Model R2: 0.6374186974488816
PCA Model RMSE: 1353769.7971816477
Random Forest R2: 0.61293485236197
Random Forest RMSE: 1398730.8456790478
```

Multi-Linear Regression Model: (Training)

- R-Squared 0.678
- Adj. R-Squared 0.670
- AIC 13310
- BIC 13350
- Number of predictors 11

PCA Transformed Multi-Linear Regression Model: (Training)

- R-Squared 0.642
- Adj. R-Squared 0.637
- AIC 13340
- BIC 13370
- Number of predictors 6

Multi-Linear Regression Model: (Testing)

- R-Squared 0.644
- RMSE 1341936.56

PCA Transformed Multi-Linear Regression Model: (Testing)

- R-Squared 0.637
- RMSE 1353769.80

Random Forest Regression Model: (Testing)

- R-Squared 0.613
- RMSE 1398730.85

Both models performed slightly better on the training data than on the testing data
R-squared in the model summary is higher than the R-Squared from the predictions

The Linear Regression model performed slightly better

The PCA transformed Linear Regression model performed more or less the same

This is because PCA is designed for maximum variance and for eliminating multi-collinearity and most importantly, dimensionality reduction

Since the indpendant variables where already uncorrelated, we are no seeing PCA

at its greatest potential

AIC and BIC for the PCA model are slightly higher than the baseline LR model, meaning that while there was a reduction of complexity, the model still fails to capture a good fit compared to the regular LR model

So while it transformed the data, it does so in a way that simplifies existing patterns meaning it's more or less the same

The RMSE is slightly higher for the PCA model meaning that it is slightly less accurate with a higher prediction error

The random forest model perfomed the poorest with an R-Squared value of 0.613 and the highest RMSE value so far

This may be due to overfitting or the fact that the data is already mostly linear, and the random forest overcomplicates that, it also may be due to it being a small dataset in general

Nevertheless it was insightful too see how all three models performed

In summary:

Multi-Linear Regression Model performs slightly better than the PCA transformed Multi-Linear Regression Model and the Random Forest Regression Model overall

Further write-up on the results in my report submission

Matthew Riddell - Machine Learning Report:

The next few pages in the combined submission file are from the Report:

Please continue reading!

Also the link to the Github where all the files can be found:

https://github.com/Matthew-Riddell/Machine-Learning-CA2.1

The Report starts now!